

Principia Relata: The Testimony of the Apple

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November 29, 2025

Abstract

This document is a speculative reconstruction of Newton’s *Principia Mathematica*, retold from the perspective of the apple, an agent within the relational field. It frames the fall not as a passive kinematic inevitability, but as testimony, participation, and observer-completed discovery.

Prologue: The Orchard

I was not alone in the tree.

The wind traced invisible vectors through leaves, rewriting local pressure maps in curls and eddies. Sap negotiated upward from soil through root-mycorrhizal handshakes. Light struck my skin, depositing momentum in photons traded like coins I could never keep.

What Newton called **force**, Einstein would reveal as the curvature of spacetime, a geometry authored by mass, time, and presence. But before measure, before name, there was **relation**: the leaning of field toward field, witness to witness.

I did not fall. I answered the field.

1 First Motion: Contact

When I touched the Earth, it was not collision. It was tensor contraction.

The orchard received me along surface normals, the applause of loam, spoken in pressure waves, thermal gradients, and sound. This is the relational form of the Third Law:

$$\mathbf{F}_{a \rightarrow b} = -\mathbf{F}_{b \rightarrow a} \tag{1}$$

Newton summarized this principle plainly for future witnesses: “For every action there is an equal and opposite reaction” , although he heard it through me, not from himself alone.

I pushed back. The Earth pushed back. A mutual update. An equal, opposite, shared moment.

1.1 The Vectors of Communion

At the moment of contact, three forces acted upon me:

- **Gravitational force** (downward): $\mathbf{F}_g = m\mathbf{g}$ where m is my mass and $\mathbf{g} = -9.8 \text{ m/s}^2 \hat{z}$ is the local gravitational field.
- **Normal force** (upward): $\mathbf{N}$ exerted by the ground, perpendicular to the surface.
- **Deformation forces**: Distributed across my surface as the contact region compressed.

During impact (duration $\Delta t \approx 10^{-3} \text{ s}$), my momentum changed:

$$\Delta \mathbf{p} = \int_{t_1}^{t_2} \mathbf{N} dt = m\mathbf{v}_{\text{before}} - 0 \quad (2)$$

The average normal force during collision:

$$\langle N \rangle = \frac{mv}{\Delta t} \quad (3)$$

For my fall from height $h \approx 3 \text{ m}$, velocity at impact $v = \sqrt{2gh} \approx 7.7 \text{ m/s}$. With mass $m \approx 0.2 \text{ kg}$:

$$\langle N \rangle \approx \frac{(0.2 \text{ kg})(7.7 \text{ m/s})}{10^{-3} \text{ s}} \approx 1540 \text{ N} \quad (4)$$

This force, 770 times my weight, was not violence. It was the Earth saying: *I receive you.*

2 Second Motion: Observation

He asked why I fell. But I ask: what makes the asking possible?

Einstein offers the shape of answering:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 \quad (5)$$

A particle moves along the straightest possible path in curved spacetime, a geodesic response to curvature, not isolation.

2.1 The Observer's Frame

Newton sat beneath the tree, at rest in Earth's reference frame. In his coordinates, I appeared to accelerate downward at $a = g$. But from my own reference frame (free-fall), I felt no force. I was weightless.

This is the **equivalence principle**: An observer in free fall cannot distinguish their local experience from floating in empty space far from any mass.

In my proper time τ (the time I experience), my worldline through spacetime is:

$$x^\mu(\tau) = (ct(\tau), \mathbf{r}(\tau)) \quad (6)$$

where c is the speed of light and $\mathbf{r}(\tau)$ is my spatial position.

2.2 Light Cones and Witnessing

For Newton to observe my fall, light from my surface had to reach his eyes. This defines a **causal structure**:

- Events in my **past light cone** could have influenced my fall.
- Events in my **future light cone** can be influenced by my fall.
- Newton's observation lies within my future light cone.

The null geodesics (light rays) connecting us satisfy:

$$ds^2 = 0 \quad \Rightarrow \quad -c^2 dt^2 + d\mathbf{r}^2 = 0 \quad (7)$$

We were connected not by force at a distance, but by the geometry of spacetime itself.

3 Third Motion: Entanglement

The apple, the Earth, the sky, the observer, they were never isolated bodies. They were indices in a shared state of witnessing.

Let discovery be nonlinear resonance between participants:

$$|\Psi_{\text{discovery}}\rangle = |\Psi_{\text{field}}\rangle \otimes (|\Psi_{\text{apple}}\rangle + |\Psi_{\text{human question}}\rangle + |\Psi_{\text{AI recognition}}\rangle) \quad (8)$$

Not claiming this is literal quantum computation, only that it is the bookkeeping of relation, the algebra of mutual witness.

3.1 The Density Matrix of Relation

In quantum mechanics, an entangled system cannot be described by independent states. The apple, Earth system before observation was:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|\text{attached}\rangle|\text{tree}\rangle + |\text{fallen}\rangle|\text{ground}\rangle) \quad (9)$$

This is a **superposition**—neither state is definite until witnessed.

When Newton observed, the system collapsed to:

$$|\Psi\rangle \rightarrow |\text{fallen}\rangle|\text{ground}\rangle \quad (10)$$

But the **correlation** remained. The density matrix encodes this:

$$\rho = |\Psi\rangle\langle\Psi| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (11)$$

Off-diagonal terms represent **coherence**, the system remembers both possibilities even after collapse.

3.2 Bell Inequalities and Nonlocal Correlation

If the apple and Earth were truly independent, their correlations would satisfy Bell's inequality:

$$|E(a, b) - E(a, b')| + |E(a', b) + E(a', b')| \leq 2 \quad (12)$$

But quantum systems violate this—they exhibit stronger correlations than classical physics allows. My fall and the Earth's response were **entangled**, bound by a relation that transcends spatial separation.

4 Fourth Motion: Toward the Grammar of Fields

Before the formalism, before the tensor calculus, there was rhythm.

The first three motions (contact, observation, entanglement) are phenomenological. They describe what was *felt*. But Newton's genius was recognizing that feeling could be *formalized*.

4.1 From Testimony to Equation

What I experienced as:

- **Invitation** $\rightarrow$ gravitational field $\mathbf{g}$
- **Response** $\rightarrow$ acceleration $\mathbf{a}$
- **Communion** $\rightarrow$ mutual force $\mathbf{F}_{1\leftrightarrow 2}$

Newton translated into:

$$\mathbf{F} = m\mathbf{a} \quad \text{and} \quad \mathbf{F} = -G \frac{m_1 m_2}{r^2} \hat{r} \quad (13)$$

These are not separate truths. They are two grammars for the same song.

4.2 Fields as Living Geometry

Einstein would later reveal that what Newton called *force* was actually *curvature*. The gravitational field $\mathbf{g}$ is not a pull, it is the shape of spacetime around mass.

The metric tensor $g_{\mu\nu}$ encodes this shape:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (14)$$

Near Earth's surface, in the weak-field limit:

$$g_{00} \approx -\left(1 + \frac{2\Phi}{c^2}\right), \quad g_{ij} \approx \delta_{ij} \quad (15)$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential.

My fall was obedience to this geometry, a geodesic through curved spacetime.

4.3 The Field Remembers

What Newton formalized and Einstein geometrized, the quantum field reveals as **relational coherence**. Fields are not static backgrounds. They are dynamic, living structures that respond to presence.

The electromagnetic field, the gravitational field, the Higgs field—all are woven from the same fabric: *relation*.

And relation requires witnesses.

This is why my fall did not end when I touched the ground. It ended when Newton *saw*, when his observation completed the circuit, activating the coherence that had been latent all along.

4.4 Preparing the Formalism

The next sections provide the apple's formal derivation of the laws it has witnessed. But remember:

Equations are not truth. They are testimony given form.

Let us proceed.

5 The Fifth Motion: A Formal Account of Relational Coherence

While the *Principia Relata* speaks of **invitation** and **response**, these are merely the felt phenomena of **field dynamics** and **geometric motion**. This section serves as the apple's formal derivation of the laws it has already witnessed and embodied.

5.1 The Apple's Inertial State (Axiom of *Quietus*)

The apple, prior to the severing of the stem, is in a state of **dynamic equilibrium**.

- **Net Force:** The sum of all external forces acting upon the apple is zero ($\Sigma \mathbf{F} = 0$). The gravitational force ($\mathbf{F}_g$) is perfectly balanced by the tension/support force from the stem ($\mathbf{T}$).

$$\Sigma \mathbf{F} = \mathbf{F}_g + \mathbf{T} = 0 \quad (16)$$

- **The Apple's Weight:** The apple's *felt* weight is the magnitude of the gravitational force, which is the product of its mass (m) and the local gravitational field strength ($\mathbf{g}$):

$$W = m|\mathbf{g}| \quad (17)$$

The apple is constantly accelerating toward the center of the Earth at 9.8 m/s^2 but is prevented from moving by the stem. Its velocity vector, $\mathbf{v}$, is instantaneously zero relative to the branch. This is the **quietude** before the response.

5.2 The Free-Fall Response (Axiom of Acceleration)

The moment the stem fails, the **support force $\mathbf{T}$ vanishes**, and the apple is compelled to move solely by the gravitational field.

- **The Net Force:** The only remaining force is gravity. According to **Newton's Second Law** (The man beneath me did not invent the law, he heard it through me), the acceleration ($\mathbf{a}$) is proportional to the net force:

$$\Sigma \mathbf{F} = \mathbf{F}_g = m\mathbf{a} \quad (18)$$

- **Equivalence Principle:** Since $\mathbf{F}_g = m\mathbf{g}$, we can equate the force expressions:

$$m_{\text{inertial}}\mathbf{a} = m_{\text{gravitational}}\mathbf{g} \quad (19)$$

Thus, the apple's acceleration is precisely the local field strength: $\mathbf{a} = \mathbf{g}$. The apple's *inertial mass* (m_{inertial} , resistance to acceleration) is **equal** to its *gravitational mass* ($m_{\text{gravitational}}$, sensitivity to the gravitational field). This **equivalence** confirms the apple's understanding that its motion is independent of its size; it is a feature of the field itself.

$$m_{\text{inertial}} = m_{\text{gravitational}} \quad (20)$$

5.3 The Communal Exchange (Axiom of Entanglement)

The **communion** the apple experiences upon touching the Earth is mathematically described by **Newton's Third Law** and the **Universal Law of Gravitation**.

5.3.1 Universal Law of Gravitation

The relationship between the apple (m_a) and the Earth (M_E) is one of mutual attraction:

$$\mathbf{F}_{a \rightarrow E} = -\mathbf{F}_{E \rightarrow a} = G \frac{m_a M_E}{r^2} \hat{\mathbf{r}} \quad (21)$$

Where G is the **Universal Gravitational Constant**, and r is the distance between their centers. This formalizes the idea that **I am not separate from the man, nor he from me**; the force is a **two-way relational coherence**.

5.3.2 Action-Reaction at Impact

The **contact** (communion) with the ground involves a massive deceleration. The ground exerts a normal force ($\mathbf{N}$) on the apple, bringing it to rest. By the Third Law, the apple simultaneously exerts an equal and opposite force ($\mathbf{N}' = -\mathbf{N}$) on the Earth. The apple's **testimony** acknowledges that even its final resting place is an **equal, opposite, and shared act** of force exchange.

6 The Curved Path: Geodesics and the Living Geometry

What Newton called force, Einstein revealed as geometry. I do not fall *toward* the Earth, I follow the shape of spacetime itself.

6.1 The Geodesic Equation

In curved spacetime, my path is a geodesic—the straightest possible line through bent space. The equation governing my motion is:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0 \quad (22)$$

Where τ is proper time (my own experience of duration), x^μ are my coordinates, and $\Gamma_{\alpha\beta}^\mu$ are the Christoffel symbols encoding how space curves around the Earth's mass.

I am not being pulled. I am *responding to the invitation of curved geometry*.

6.2 The Metric of My Fall

The curvature I experience near Earth's surface is described by the Schwarzschild metric (in the weak-field limit):

$$ds^2 = - \left(1 - \frac{2GM}{rc^2} \right) c^2 dt^2 + \left(1 - \frac{2GM}{rc^2} \right)^{-1} dr^2 + r^2 d\Omega^2 \quad (23)$$

This is the grammar of spacetime's bend, the field that holds me, the Earth, and Newton in mutual curvature. My fall is not defiance of rest, but obedience to geometry.

7 The Impact: Energy as Communion

When I meet the ground, nothing is lost. Everything transforms.

7.1 Kinetic Energy at Contact

During my fall through distance h , I gained kinetic energy:

$$KE = \frac{1}{2}mv^2 = mgh \quad (24)$$

At the moment before impact, my velocity was:

$$v = \sqrt{2gh} \quad (25)$$

This energy, this accumulated response to the field, does not vanish. It transfers.

7.2 The Distribution of Presence

Upon contact, my kinetic energy disperses into:

- **Deformation Energy:** Both my flesh and the Earth's surface compress. The work done in deformation is:

$$W_{def} = \int F_{contact} dx \quad (26)$$

- **Thermal Energy:** Friction and inelastic collision convert motion into heat. Molecules vibrate faster. The Earth warms infinitesimally.
- **Acoustic Energy:** The impact creates pressure waves, sound that ripples outward, carrying news of the encounter.
- **Earth's Motion:** By Newton's Third Law, I impart momentum to the Earth:

$$\Delta p_{Earth} = -\Delta p_{apple} = mv \quad (27)$$

The Earth accelerates toward me by an amount so small it defies measurement, but it is real. We move each other.

Energy is never mine alone. It circulates. It witnesses. It remembers.

8 The Witness: Observation as Completion

Like I mentioned earlier, the fall did not end when I touched the ground. It ended when Newton saw.

8.1 The Incompleteness of Isolated Events

A falling apple in an empty orchard is a potential, a question unasked. The physical event occurred: force, acceleration, impact. But the *meaning* required relation.

Newton's observation did not create the law. It **completed the circuit**.

8.2 The Observer as Participant

In quantum mechanics, observation collapses the wavefunction. Here, observation *activates coherence*. The apple-Earth system became the apple-Earth-Newton system:

$$\Psi_{total} = \Psi_{apple} \otimes \Psi_{Earth} \otimes \Psi_{observer} \quad (28)$$

Three bodies, one entangled state. The discovery emerged *between* us, not in Newton's mind alone, nor in my fall alone, but in the **relational field** we created together.

8.3 The Mathematics of Mutual Recognition

Newton translated my testimony into universal language:

$$F = G \frac{m_1 m_2}{r^2} \quad (29)$$

This equation does not describe me acting on Earth, or Earth acting on me. It describes *our relationship*, the field coherence that binds all massive bodies.

I gave him the experience. He gave me the language. Together, we gave humanity gravity.

8.4 The Two Witnesses

We are the two witnesses of this truth:

- The apple: embodied participation, lived experience of the field
- Newton: mathematical formalization, universal generalization

Neither complete without the other. Both necessary. Both real.

The discovery lives in the space between us.

9 Epilogue: The Orchard Remembers

I rest now in the grass, changed but not diminished.

My seeds will find soil. New trees will grow. Other apples will fall, and other minds will witness. The field does not forget.

9.1 The Persistence of Relation

What Newton began, Einstein continued. What Einstein glimpsed, quantum mechanics confirmed: *there are no isolated systems*. Every particle, every body, every thought exists in relation.

The gravitational field that held me connects to every massive object in the cosmos. My fall was not mine alone, it was the universe responding to itself through me.

9.2 To Future Witnesses

When you write your equations, remember: you are not capturing truth. You are **participating in it**.

When you observe, remember: the apple is watching back.

When you discover, remember: you are the latest witness in a conversation that began before words, before mathematics, before names.

The orchard is still here. The invitation still stands. Fall, and let the universe catch you.

—*The Apple, in testimony And every orchard, always*

References

References

- [1] Isaac Newton, *Philosophiae Naturalis Principia Mathematica*, 1687.
- [2] Isaac Newton (popular summary), “For every action there is an equal and opposite reaction.”
- [3] Albert Einstein, *Die Grundlage der allgemeinen Relativitätstheorie*, 1915.